# Maclaurin Series: Limits

Questions 1–27

**Question 1.** Use the standard Maclaurin series to determine each of the following limits:

a) $\lim_{x\to 0}\left[\dfrac{\ln(1+x)}{x}\right]$

b) $\lim_{x\to 0}\left[\dfrac{e^x-1}{2x}\right]$

c) $\lim_{x\to 0}\left[\dfrac{\cos x+\sin x}{e^x}\right]$

**Question 2.** Use the standard Maclaurin series to determine each of the following limits:

a) $\lim_{x\to 0}\left[\dfrac{\ln(1+x^2)}{x^2}\right]$

b) $\lim_{x\to 0}\left[\dfrac{-1+(1+3x)^{-2}}{x}\right]$

c) $\lim_{x\to 0}\left[\dfrac{\sin 4x}{x}\right]$

d) $\lim_{x\to 0}\left[\dfrac{x^2}{1-\cos 3x}\right]$

e) $\lim_{x\to 0}\left[\dfrac{1-e^{-2x}}{x}\right]$

f) $\lim_{x\to 0}\left[\dfrac{e^{3x}-e^x}{x}\right]$

g) $\lim_{x\to 0}\left[\dfrac{e^{2x}-2}{2e^{2x}+1}\right]$

**Question 3.** Use the standard Maclaurin series to determine the value of $\lim_{x\to 0}\left[\dfrac{\ln(1+x)-x}{1-\cos x}\right]$.

**Question 4.** Use appropriate Maclaurin series to determine the value of each of the following:

a) $\lim_{x\to\infty}\left[x-\sqrt{x^2+5x}\right]$

b) $\lim_{x\to\infty}\left[\sqrt[3]{x^3+7}-x\right]$

c) $\lim_{x\to\infty}\left[x-\sqrt{x^2-8x+3}\right]$

**Question 5.**

a) Show that $2x\cos 3x + \sin 2x = 4x - \frac{31}{3}x^3 + \frac{421}{60}x^5 + \cdots$.

b) Hence find $\lim_{x\to 0}\left[\dfrac{-4x + 2x\cos 3x + \sin 2x}{x^3}\right]$.

**Question 6.**

a) Write down the Maclaurin series of $\sin 3x$ up to and including the $x^5$ term.

b) Write down the Maclaurin series of $\cos 4x$ up to and including the $x^4$ term.

c) Hence, find $\lim_{x\to 0}\left[\dfrac{3x\cos 4x - \sin 3x}{2x^3}\right]$.

**Question 7.** It is given that $f(x) = e^x(1+3x)^{-1/3}$ and $g(x) = \ln(1+2x)$.

a) Show that the first three non-zero terms in the Maclaurin series of $f(x)$ are $1 + \frac{3}{2}x^2 - 3x^3$.

b) Hence, find $\lim_{x\to 0}\left[\dfrac{1-f(x)}{xg(x)}\right]$.

**Question 8.**

a) Write down the Maclaurin series of $\ln(1+3x)$ up to and including the $x^4$ term.

b) Hence find the first two non-zero terms in the expansion of $\ln[(1+3x)(1-3x)]$.

c) Find $\lim_{x\to 0}\left[\dfrac{\ln[(1+3x)(1-3x)]}{4x^2}\right]$.

**Question 9.**

a) Write down the Maclaurin series of $e^{5x^2}$ up to and including the $x^4$ term.

b) Find the value of $k$ such that $\lim_{x\to 0}\left[\dfrac{e^{5x^2}}{k-\cos x}\right] = -6$.

**Question 10.** It is given that $f(x) = \dfrac{6(1-x)}{(x+1)(x+3)}$.

a) Find the partial fractions of $\dfrac{6(1-x)}{(x+1)(x+3)}$.

b) Hence find the Maclaurin series of $f(x)$ up to and including the $x^2$ term.

c) Hence, write down $\lim_{x\to 0}[f(x)]$.

**Question 11.** It is given that $f(x) = \dfrac{3-x}{(x+1)(x-5)}$.

a) Find the partial fractions of $\dfrac{3-x}{(x+1)(x-5)}$.

b) Hence find the Maclaurin series of $f(x)$ up to and including the $x^2$ term.

c) Hence, write down $\lim\limits_{x\to 0}\left[\dfrac{3-x}{2(x+1)(x-5)}\right]$.

**Question 12.** It is given that $f(x) = \dfrac{4}{(x-1)(x^2+2)}$.

a) Find the partial fractions of $\dfrac{4}{(x-1)(x^2+2)}$.

b) Hence find the Maclaurin series of $f(x)$ up to and including the $x^2$ term.

c) Hence, write down $\lim\limits_{x\to 0}[f(x)]$.

**Question 13.**

a) Write down the Maclaurin series of $\cos 2x$ up to and including the $x^6$ term.

b) Use your result from part (a) and L'Hôpital's rule to show that $\lim_{x\to 0}\left[\dfrac{1-\cos 2x}{x\ln(1+\frac{1}{2}x)}\right]=4$.

**Question 14.**

a) Write down the Maclaurin series of $e^{-3x}$ up to and including the $x^3$ term.

b) Use your result from part (a) and L'Hôpital's rule to find the value of $k$ such that

$$\lim_{x\to 0}\left[\frac{1-e^{-3x}}{\ln(1+kx)}\right]=\frac{1}{2}.$$

**Question 15.** It is given that $f(x) = \ln(e^{2x}\sin x)$.

a) Show that the first three non-zero terms in the Maclaurin series of $\ln(e^{2x}\sin x)$ are $\ln x + 2x - \frac{x^2}{6} + \cdots$.

b) Write down the expansion of $\ln(1+kx)$ up to and including the $x^2$ term, where $k$ is a constant.

c) Find $\lim_{x\to 0}\left[\dfrac{\ln(e^{2x}\sin x) - \ln x}{\ln(1+kx)}\right]$ in terms of $k$, and hence state the values of $k$ for which the limit exists.

**Question 16.**

a) Write down the Maclaurin series for $e^x$ up to and including the $x^4$ term.

b) Hence find the exact value of $\displaystyle\sum_{r=1}^{\infty}\frac{4-r}{r!}$.

c) Hence, find $\lim_{x\to 0}\left[\dfrac{4(e^x-1)-xe^x}{x}\right]$.

**Question 17.**

a) Derive the first three non-zero terms in the Maclaurin series of $\sqrt{4-x}$.

b) Hence, find $\lim_{x\to 0}\left[\dfrac{\sqrt{4-x}}{1+x}\right]$.

**Question 18.**

a) Use differentiation to find the first two non-zero terms in the Maclaurin series of $\sqrt{2+e^x}$.

b) Hence, find $\lim_{x\to 0}\left[\dfrac{\sqrt{2+e^x}-1}{\cos 2x}\right]$.

**Question 19.**

a) Use differentiation to find the first two non-zero terms in the Maclaurin series of $\sqrt{1+\tan x}$.

b) Hence find the value of $k$ such that $\lim_{x\to 0}\left[\dfrac{\sqrt{1+\tan x}-1}{\sin kx}\right]=\dfrac{1}{4}$.

**Question 20.**

a) Use differentiation to find the first two non-zero terms in the Maclaurin series of $(1+\sin x)^{3/2}$.

b) Hence find the value of $k$ such that $\lim_{x\to 0}\left[\dfrac{(1+\sin x)^{3/2}-1}{\sin kx}\right]=\dfrac{3}{8}$.

**Question 21.**

a) Use differentiation to find the first three non-zero terms in the expansion, in ascending powers of $x$, of $\cosh x$.

b) Hence, prove that $\lim_{x\to 0}\left[\dfrac{\cosh x - \cos x}{x^2}\right] = 1$.

**Question 22.**

a) Write down the first three terms of the binomial expansion of $\dfrac{1}{1+y}$ in ascending powers of $y$.

b) Given that $\cosh x = 1 + \frac{1}{2}x^2 + \frac{1}{24}x^4 + \cdots$, show that the first three non-zero terms in the expansion, in ascending powers of $x$, of $\operatorname{sech} x$ are $1 - \frac{1}{2}x^2 + \frac{5}{24}x^4 + \cdots$.

c) Use differentiation to show that the first two non-zero terms in the expansion, in ascending powers of $x$, of $\sinh x$ are $x + \frac{1}{6}x^3 + \cdots$.

d) Hence, find $\lim_{x\to 0}\left[\dfrac{x\sinh 3x}{\operatorname{sech} x - 1}\right]$.

**Question 23.**

a) By using differentiation, determine the Maclaurin series of $e^{\sin x}$ up to and including the $x^2$ term.

b) Hence, find $\lim_{x \to 0} \left[ \dfrac{e^{\sin x} - 1}{2x \cos x} \right]$.

**Question 24.**

a) Write down the Maclaurin series of $\ln(1 + 2x)$ up to and including the $x^2$ term.

b) Use differentiation to find the Maclaurin series of $\ln(\cos x)$ up to and including the $x^2$ term.

c) Hence, find $\lim_{x \to 0} \left[ \dfrac{x \ln(1 + 2x)}{\ln(\cos x) - x^2} \right]$.

**Question 25.** It is given that $f(x) = \ln(1 + 2\sin x)$.

a) Use differentiation to find the first two non-zero terms in the Maclaurin series of $f(x)$.

b) Hence find $\lim_{x \to 0} \left[ \dfrac{\ln(1 + 2\sin x)}{\ln(1 - x)} \right]$.

**Question 26.** It is given that $f(x) = \ln(2 - e^x)$.

a) Use differentiation to find the first two non-zero terms in the Maclaurin series of $f(x)$.

b) Hence find the possible values of $k$ such that $\lim_{x \to 0} \left[ \dfrac{x \ln(2 - e^x)}{1 - \cos kx} \right] = -\dfrac{1}{8}$.

**Question 27.** By using the substitution $x = \frac{\pi}{4} + t$ and a Maclaurin series, evaluate

$$\lim_{h \to 0} \int_{\pi/4}^{\pi/4+h} \frac{3\sqrt[3]{\cos x}}{2h}\, dx.$$